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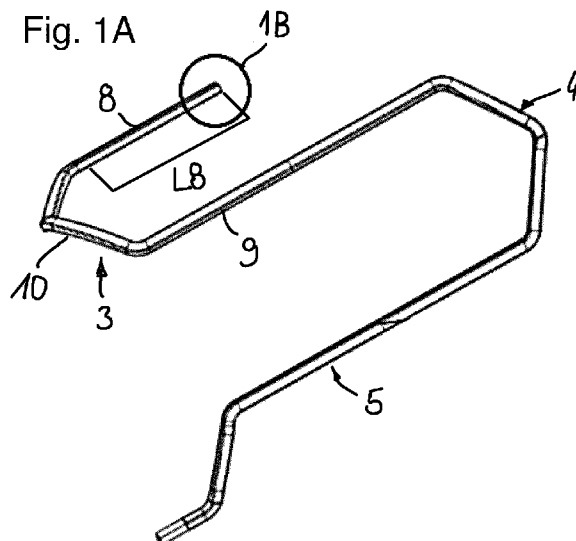
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(54) Title: CONDUCTOR FOR AN ELECTRIC MACHINE



(57) Abstract: The invention relates to a conductor (2) for an electric machine, comprising: a first winding element (3) made of an electrically conductive material and having a channel (7) through which a coolant can flow, wherein the first winding element (3) is integrally formed with a first straight portion (8), a reverse portion (10) and a second straight portion (9) parallel to the first straight portion (8), wherein the first straight portion (8) has a first end section (12) and the second straight portion (9) has a second end section (13); a hollow connecting member (6) made of electrically conductive material; wherein the hollow connecting member (6) and the first end section (12) of the winding element (3) are axially insertable into each other to form an axially overlapping electrical and fluidic connection. The invention further relates to an electric machine (25) with such a conductor (2).

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Conductor for an electric machine

Description

The present invention relates to a hollow winding element for an electric machine. The electric machine can be any type of motor with a winding stator. In particular, a permanent magnet synchronous motor, an induction motor, an excited rotor synchronous motor or a switched reluctance motor. An electric motor is an electric machine which typically comprises a stator and a rotor. The stator carries windings connected to an alternating current (AC) power supply to produce a rotating magnetic field.

10 Document US 2021/0249926 A1 discloses a coil bobbin attached to a stator core of a distributed winding radial gap-type rotating electric machine, and including a teeth holding portion and a slot insulator which are made of an insulator. The teeth holding portion has: a first wall surface that covers a first circumferential side surface of a tooth of the stator core; a second wall surface that covers at least a part of a second circumferential side surface of the tooth; and a third wall surface that covers both side surfaces in an axial direction of the tooth. The slot insulator is formed integrally with the first wall surface of the teeth holding portion, and has a plurality of through-holes extending in the axial direction and arrayed in a radial direction.

20 From DE 10 2020 114 683 A1, an active part for an electrical machine is known that comprises a current-conductive winding for generating a magnetic flux and an active part iron for holding the current-conductive winding for conducting the magnetic flux of the winding. The active part iron has a plurality of axially extending grooves over the circumference, in which the winding conductors are arranged. The winding conductors are formed as hollow conductors, thereby forming a cooling channel for a coolant for cooling the active part. The winding is designed as a wave winding with one-piece winding conductors.

From document US 2021 0091619 A1, corresponding to WO 2019 038139 A1, a winding piece is known which is to be inserted into one or more grooves of an electrical machine. The winding piece is in the shape of a hairpin, which are inserted into the grooves of the electrical machine. The winding piece comprises two limbs and a curved transition region, which connects the two limbs to each other. The limbs each have a first portion with a profile adjusted to the cross-section of the groove, and an end portion with a round profile. The winding piece is formed as a hollow conductor, which has a penetrating channel, through which a coolant can be led.

There are known some direct oil cooling solutions to cool down the winding of a PMSM. Normally, these solutions are focused on end windings, while the active winding inside the stator stack remains relatively far from any cooling sources. Thus, heat is higher on the active winding, with the conductor layer closest to the airgap contains the hotspot. Furthermore, there are some solutions for active winding cooling, but they all remain far from the hotspot though.

It is an object of the present invention to propose a conductor for an electric machine with an effective cooling structure that can be produced easily and cost effectively. The object is further to propose an electric machine with such a conductor.

According to the invention, a conductor to be inserted into a slot of an electrical machine is proposed, comprising: a first winding element made of an electrically conductive material as a hollow conductor with a channel through which a coolant can flow, wherein the first winding element includes integrally formed a first straight portion, reverse portion and a second straight portion parallel to the first straight portion, wherein the first straight portion has a first end section and the second straight portion has a second end section; a hollow connecting member made of electrically conductive material; wherein the hollow connecting member and the first end section of the winding element are axially insertable into each other to form an axially overlapping electrical and fluidic connection. The first straight portion, reverse portion and second straight portion form one piece, i.e. they are portions of the same winding element. The winding element can be described to be hairpin-shaped.

An advantage of the conductor is that the hollow winding elements and hollow connecting members can serve as electric conductors with a coolant flow circulating inside them. The cooling fluid can be an oil, or oil based fluid, for example. A conductor composed of a plurality of winding elements and hollow members connecting them, can be
5 fed with a cooling fluid so that the winding can be cooled down directly in the source of the heat losses that it generates. This is a very effective way to cool the winding as the loss generation center point is tackled.

A conductor is made of a plurality of single winding elements which can also be referred
10 to as conductor elements or hair pin elements. Two winding elements are connected to each other via a respective connecting member. According to a first option, the connecting member may be a separate part that is introduced into both adjoining winding elements so as to obtain a mechanic, electrical and fluidic connection. The connection can be established very easy by a simple plug-in movement in axial direction. No further
15 connecting means, such as welding, are necessary. The two winding elements are connected to each other on their straight portion with a connecting member that can also be referred to as tube or sleeve member. According to a second option, the connecting member may also be a portion of the second winding element. In this case, the first winding element and the second winding element are directly plugged into one
20 another. The two connected winding elements are arranged in the same slot of the stator and may have the same layer span and/or bended geometry.

Another advantage is that the tube member and the winding element(s) can be easily
25 connected with each other by a simple plug-in connection. As mentioned, no welding is required, thus reducing process and time costs. Furthermore, since the connection is arranged within the slot of the electric machine, the reverse portions of the winding elements can be shaped according to the requirements, with the overall length of the machine being particularly small.

30 The hollow connecting member can be connected to the first end section of the winding element with an interference fit. In this manner, a particularly secure mechanical, electrical and fluidic connection is achieved. However, it is understood that the connecting member and winding element could also be configured with a transition fit, or small play fit. An end portion of the tube member and/or the end face of the winding member

can have a conical chamfer for easy insertion of the tube member into the bore of the winding member. The axial length of the tube member can be at least 0.1 times, in particular at least 0.2 times the length of the first straight portion of the winding element. This contributes to a safe connection.

5

According to an embodiment, the wall thickness of the tube member can be smaller than the wall thickness of the winding element.

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The end section of the winding element can have a connecting bore that extends from the end face of the winding element to a stop face. The bore forms an interior hole that can be machined into the end section of the winding element so as to have a greater cross section than the channel of the winding element. Or, in other words, the inner diameter of the bore can be greater than the inner diameter of the channel of the winding element at a section adjoining the end section. The connecting tube can be inserted into the bore until it axially abuts the stop face of the winding element. Then, with its other end, the connecting tube can be inserted into the second winding element.

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The outer diameter of the connecting member can be greater than the inner diameter of the channel of the winding element. The connecting member has a smallest inner diameter which, for example, can be greater than 0.9 times and/or smaller than 1.1 times a smallest inner diameter of the channel of the winding element. In this way, any change in cross section of the channel along the connection, and thus any pressure jump, is small. This contributes to a good flow of the cooling fluid through the channel. Preferably, the cross section and/or diameter of the channel is substantially constant along the connecting region between the tube and the winding elements so that a smooth flow is achieved.

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The connecting member and/or the winding element can be produced from copper, aluminium, stainless steel or an alloy comprising at least one of copper, aluminium and stainless steel. The materials of the connecting member and the winding element are preferably selected such that a difference between the thermal expansion coefficient of the material of the connecting member and the thermal expansion coefficient of the material of the winding element is smaller than 10 % of the friction coefficient of any one of the materials.

According to a first embodiment, a second winding element can be provided with the same structure as the first winding element, wherein the first winding element and the second winding element are connected with each other via the hollow connecting member. On one side, the connecting member is axially inserted into the first end section of the first winding element, and on the other side it is axially inserted into the second end section of the second winding element. The conductor can be composed of a plurality of first and second winding elements that are connected to each other via a plurality of connecting members, wherein the conductor forms a continuous cooling channel with an inlet and an outlet.

According to a second embodiment, the channel can comprise a plurality for first and second winding elements, with straight intermediate elements arranged therebetween. In this case, a first winding element can be connected via a hollow connecting member with a straight intermediate element, with one side of the hollow connecting member axially inserted into an end section of the first winding element and the other side of the hollow connecting member axially inserted into the straight intermediate element. The other end of the straight intermediate element is connected via another hollow connecting member to another winding element, and so on, to form a continuous conductor. Further details regarding the connections between the winding elements and the straight elements via the hollow connecting members can be configured as described above.

A cooling fluid can be supplied to the circuit on a terminal rack. The terminal rack can have blocks, instead of plates for terminal, which allows not only the inverter terminal connection but as well oil inlets. On the neutral points of the winding arrangement, the conductors can end in an open terminal, squirting the oil over the end windings and to the motor oil pool. Hydraulically, all parallels of all phases can be connected in parallel.

The above object is further achieved with an electric machine comprising a stator and a rotor, wherein the stator comprises a plurality of longitudinal slots distributed over the circumference, with a plurality of conductors being arranged in each one of the longitudinal slots of the stator for generating an electromagnetic field, wherein at least two of the plurality of conductors is configured according to any one of the embodiments

described above. The slots extend in longitudinal, i.e. axial direction of the stator and can also be designated as grooves.

The advantages of the electric machine correspond the advantages described above
5 in connection with the conductor. A direct cooling of the conductor core and/or heat source is achieved. This allows that the electric machine, assuming the same current density, can be operated at a lower temperature, thus reducing the deterioration of the insulation system. Accordingly, higher torque densities of the electric machine can be obtained for the same current densities. To achieve a maximum operating condition,
10 the current of the cooling fluid can be increased, hence improving the continuous operation torque and speed envelopes, as well as higher peak torque and power.

According to an embodiment, the stator can comprise at least four conductors in each slot, wherein at least two of the plurality of conductors are configured to be solid con-
15 ductors. In other words, the electric machine can be provided with a partial number of hollow conductors configured according to the invention and a partial number of solid conductors. Preferably, at least one hollow conductor is arranged in each one of the slots of the stator. The connecting tubes of the plurality of conductors are preferably arranged in a center plane arranged between a first end face and an opposite second
20 end face of the stator.

Preferred embodiments are described below with reference to the drawings. Herein

- Figure 1A shows a conductor portion according to the invention in a perspective
25 view;
- Figure 1B shows two winding elements with a connecting member of Figure 1A in an exploded view;
- Figure 1C shows the end section of a winding element of the conductor portion of Figure 1A;
- 30 Figure 1D shows the connecting region between two winding elements of the conductor portion of Figure 1A;
- Figure 2A shows a connecting member of the conductor portion of Figure 1A as a detail in an axial view;

- Figure 2B shows the connecting member of Figure 2A in a side view;
- 5 Figure 3A shows the conductor portion of Figure 1A in an axially exploded view with a stator of an electric machine;
- Figure 3B shows the stator and conductor portion of Figure 3A in an axial view;
- 10 Figure 4 shows a conductor according to the invention with a plurality of winding elements connected to each other to form a plurality of straight portions and reverse portions over the circumference;
- Figure 5A shows the conductor of Figure 4 in a three dimensional exploded view with a stator of an electric machine;
- 15 Figure 5B shows the inlet and outlet of the conductor of Figures 4 and 5A as a detail;
- Figure 5C shows the stator and conductor of Figure 5A in a mounted condition in a three dimensional view;
- 20 Figure 6 shows an electric machine according to the invention with a plurality of conductors according to the invention in an axial view;
- Figure 7A shows a conductor portions according to the invention in second embodiment in a perspective exploded view;
- 25 Figure 7B shows a group of conductor portions in a perspective exploded view;
- Figure 7C shows conductor with a plurality of first and second winding elements and intermediate elements in a perspective exploded view;
- Figure 8A shows a stator with plurality of conductor portions according to Figure 7B in a perspective exploded view;
- 30 Figure 8B shows the stator with the plurality of conductor portions according to Figure 8A in a longitudinal sectional view;
- Figure 8C shows the detail 8C of Figure 8B in a larger view.

Figures 1A to 1D, which will be jointly described below, show a portion of a conductor 2 according to the invention. A conductor 2 is to be mounted in a plurality of circumferentially distributed slots of a stator of an electric machine (not shown here).

5 The conductor 2 comprises a plurality of winding elements 3, 4, 5 that are connected with each other via connecting members 6. A connecting member 6 is shown as a detail in Figures 2A and 2B and can also be referred to as tube or sleeve member. The winding elements 3, 4, 5 and connecting members 6 are made from an electrically conductive material. In a mounted condition, they form the hollow conductor 2 with a
10 channel 7 through which a coolant can flow. Thus, the conductor 2 has two functions, namely conducting electrical current and conducting coolant.

The winding elements 3, 4 can be made from copper, aluminium, stainless steel or an alloy comprising at least one of copper, aluminium and stainless steel. The same applies for the material of the connecting members 6. The materials of the connecting
15 member 6 and the winding element 3, 4 are preferably selected such that a difference between the thermal expansion coefficient of the material of the connecting members and the thermal expansion coefficient of the material of the winding elements is smaller than 10 % of the friction coefficient of the materials of one of them.

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As a representative of several winding elements 3, 4 for the stator, one winding element 3 is described in greater detail. A winding element 3 is hair-pin shaped, including a first straight portion 8, a second straight portion 9 parallel to the first straight portion, and a reverse portion 10 connecting the first and second straight portions 8, 9 with
25 each other. As can be seen in Figures 1B and 1C, the first and second straight portions 8, 9 have a respective first and second end section 12, 13, into which a respective connecting member 6 can be axially inserted so as to connect to winding elements 3, 4 with each other. The connecting member 6 can be connected to the first and second winding elements 3, 4 with an interference fit. For easily introducing the connecting
30 member 6 into the respective winding element 3, 4, any one of said elements may optionally have a conical chamfer. In the mounted condition, as shown in Figure 1D, an axially overlapping mechanical, electrical and fluidic connection is formed by the elements 3, 4, 6. The axial length L7 of the connecting member 6 can be for example at least 0.1 times, in particular at least 0.2 times the length L8 of the first straight portion

8 of the winding element 3. Thus, a secure axial overlapping connection is obtained. The wall thickness of the connecting member 6 can be smaller than the wall thickness of the winding elements 3, 4, as can be seen in Figure 1D.

5 The end section 12 of the winding element 3 has a connecting bore 14 that extends from the end face 15 of the winding element to a stop face 16. The bore 14 forms an interior hole that can be machined into the end section of the winding element 3 so as to have a greater cross section and/or diameter d_{14} than the cross section and/or diameter d_7 of the channel 7 of the winding element. The stop face 16 can serve as a
10 stop for the connecting member 6 to be inserted into the bore 14. The other end of the connecting member 6 is inserted into the bore of the adjacent winding element 4.

As can be seen in Figure 1D, the outer diameter D_6 of the connecting member 6 is greater than the inner diameter d_7 of the channel of the winding elements 3, 4. The
15 inner diameter d_6 of the connecting member 6 substantially corresponds with the inner diameter d_7 of the channel 7 of the winding elements, so that a good cooling flow is achieved without pressure drops. A ratio between the inner diameter d_6 of the connecting member 6 and the inner diameter d_7 of the channel 7 can be for example between 0.9 and 1.1, i.e. $0.9 < d_6/d_7 < 1.1$.

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Figures 3A and 3B show the portion of the conductor 2 of Figure 1A with the winding elements 3, 4, 5 together with a stator core 20 of an electric machine. The stator core 20 is generally ring-shaped around a stator axis A_{20} , and comprises a plurality of slots 21 being distributed on an inner circumferential face 24 around the axis A_{20} . The stator
25 core 20 can be produced, for example, by laminating a plurality of electromagnetic steel sheets or soft magnetic materials. A complete conductor 2 being composed of a plurality of winding elements 3, 4, 5 and constituting one electric and fluidic channel is shown in Figure 4. The conductor 2 shown here comprises 16 pairs of straight portions (8, 9) and reverse portions (10, 11) forming a meander-like structure over the circum-
30 ference. The straight portions are to be arranged in respective slots 21 of the stator core 20, wherein in each slot a plurality of straight portions is received. The reverse portions 10, 11 are arranged axially adjacent to the stator core 20 and span a plurality of slots in circumferential direction. As can be seen in Figure 4, each conductor 2 has

an inlet 22 and an outlet 23 so that a cooling fluid can be supplied through the conductor circuit. The cooling fluid can be supplied to the circuit on a terminal rack (not shown).

Figures 5A, 5B and 5C show the conductor 2 of Figure 4 in combination with a stator core 20. The complete winding structure to constitute the complete stator of an electric machine can have 6 conductors 2 of the type shown in Figure 4. However, it is understood that this number is only exemplary, and depending of the number of slots arranged over the circumference, and the number of conducting channels in each slots, the number of conductors may vary accordingly.

An electric machine 25 according to the invention is shown in Figure 6. The electric machine 25 comprises a stator 26 and a rotor 27 arranged coaxially thereto. The stator 26 includes the stator core 20 and the windings 28 composed of a plurality of conductors 2 according to the invention. The electric machine 25 is configured in the form of a permanent magnet synchronous machine (PMSM). The windings 28 are connected to an AC power supply to produce a rotating magnetic field. The rotor 27 comprises a rotor member 29 and a plurality of permanent magnets 30 embedded therein to create a constant magnetic field. In an activated condition, the rotor 27 is rotated about the axis of rotation A27 to drive a drive shaft 31 connected to the rotor member 29.

In this embodiment, the stator core 20 comprises a number of 48 slots 21 over the circumference, with a number of four conductor lines arranged in each slot 21. However, it is understood that these numbers are only exemplary, with stators with other numbers of slots and conductor lines being possible which can be selected according to the technical requirements.

The electric machine 25 including conductors 2 according to the invention is advantageous in that the hollow winding elements 3, 4 and connecting members 6 can serve at the same time as electric conductors and cooling conductors. The conductors 2 can be fed with a cooling fluid so that the windings 28 can be cooled down directly where the heat is generated. The coolant can be fed by a high voltage connector and can exit at a neutral phase of the electric machine 25. The coolant can either be collected or extracted to the heat exchanger or it may be collected in a carter inside the electric machine, depending whether a wet or dry motor design is provided. The coolant may

be fed on a given parallel winding. The coolant feeding may be implement at first, second, third or fourth parallel indistinctively for all the phases in the motor, ensuring a balanced neutral voltage.

5 Further advantages and special features are that the structure and construction is easy which is well compatible with mass production and thus very cost effective. The shaped hollow winding elements 3, 4 are connected both physically and electrically by means of the connecting members 6 that can robustly transfer the flow of coolant and electric current. The winding elements 3, 4 may touch axially to conduct current, in order to
10 form a supplementary current path to the channel cross section. Other shapes may be considered in order to maximize current transfer path, for example a conical shape. As mentioned above, the connecting member 6 can be installed in a press fit configuration with the two hairpin halves, thus ensuring interference and aiming for perfect contact between the conductors 2 and the connecting sleeve. The manufacturing process is defined to allow easy assembly. The connecting members 6 can be located in
15 the middle of the stator 25 or in a different position close to the end windings as required. In a modified embodiment, additional connecting sleeves may be implemented to apply existing winding elements 3, 4 forming equipment for different stack length. This may require for example two connections in the straight side of the winding elements 3, 4.
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Figure 7A shows a portion of a conductor 2 according to the invention in a second embodiment. The conductor portion widely corresponds to the embodiment according to Figures 1 to 5 to which description it is hereby referred to with regard to similarities.
25 The same or corresponding details are provided with the same reference signs as in the above Figures.

The conductor 2 comprises a plurality of winding elements 3, 4 that are connected with each other via an intermediate element 19 by means of connecting members 6, 6'. The
30 winding elements 3, 4, the intermediate element 19 and the connecting members 6, 6' are made from an electrically conductive material. In a mounted condition, they form a hollow conductor 2 with a channel 7 through which a coolant can flow.

A characterising feature of the embodiment according to Figure 7A is that the conductor does not only include first and second winding elements 3, 4, but also includes straight intermediate elements 19 to be inserted in the slots of the stator. Accordingly, the straight portions 8, 9 of the first and second winding elements 3, 4 are shorter than
5 in the first embodiment. Besides this, the structure and design of the winding elements 3, 4 corresponds to the above described embodiment according to Figures 1 to 5. An intermediate element 19 comprises a first end section 29 to be connected with an end section of a first winding element 3 via a first connecting member 6, and a second end section 29' to be connected with an end section of a second winding element 4 via a
10 second connecting member 6'. The connection between a winding element 3, 4 and the intermediate element 19 via a connecting member 6, 6' is effected as described with respect to Figures 1 to 5 to which it is referred for brevity.

Figure 7B shows a group of conductor portions which are to be connected to further
15 conductor portions (not shown) to jointly form one single conductor for an electric machine.

Figure 7C shows a conductor including a plurality of conductor portions in an exploded view. The conductor 2 is hollow and forms a channel 7 through which a coolant can
20 flow. As in the above embodiment, the conductor 2 has two functions, namely conducting electrical current and conducting coolant. The conductor 2 has an inlet 22 and an outlet 23 so that a cooling fluid can be supplied through the conductor circuit.

Figures 8A to 8C show the conductor 2 of Figure 7C with the winding elements 3, 4
25 and intermediate elements 19, together with a stator core 20 of an electric machine. The stator core 20 is configured as described in the first embodiment to which description it is hereby referred, with the same or corresponding components being provided with the same reference signs. The conductor 2 shown here comprises 16 pairs of straight intermediate elements 11 which are connected with respective winding ele-
30 ments 3, 4 to form one channel 2 with a meander-like structure over the circumference. The straight intermediate elements 11 are arranged in respective slots 21 of the stator core 20, wherein a plurality of intermediate elements 11 is received in each slot 21. As can be seen in particular in Figures 8B and 8C, the intermediate elements 11 are axially longer than the axial extension of the stator core 20 and. The ends 34, 34' of the

straight elements protrude just beyond the end face 32, 33 of the stator core 20, in particular by less than half the length L6 of the connecting member 6, 6'.

The winding elements 3, 4 are arranged axially adjacent to the stator core 20 and span a plurality of slots 21 in circumferential direction. The inlet 22 and outlet 23 of the conductor 2 can be seen in Figure 8A. The cooling fluid can be supplied to the circuit on a terminal rack (not shown). For simplicity, Figures 8A and 8B show only one channel 2. However, it is understood that in a completely mounted condition the stator 25 includes a plurality of channels 2, as described for example in connection with Figure 6.

Reference Numerals

2	conductor
3	winding element
4	winding element
5	winding element
6	connecting member
7	channel
8	straight portion
9	straight portion
10	reverse portion
11	reverse portion
12	end section
13	end section
14	bore
15	end face
16	stop face
17	end face
18	stop face
20	stator core
19	intermediate element
21	slots
22	inlet
23	outlet
24	inner face
25	electric machine
26	stator
27	rotor
28	windings
29, 29'	end section
30	permanent magnets
31	drive shaft
32	end face

33 end face

34, 34' end

A axis

d, D diameter

L length

Claims

1. Conductor to be inserted into a slot of an electrical machine, comprising:
a first winding element (3) made of an electrically conductive material as a hollow conductor with a channel (7) through which a coolant can flow, wherein the first winding element (3) is integrally formed having a first straight portion (8), a reverse portion (10) and a second straight portion (9) parallel to the first straight portion (8), wherein the first straight portion (8) has a first end section (12) and the second straight portion (9) has a second end section (13), and
a hollow connecting member (6) made of electrically conductive material,

characterised in that

the hollow connecting member (6) and the first end section (12) of the winding element (3) are axially insertable into each other to form an axially overlapping electrical and fluidic connection.
2. Conductor according to claim 1,
characterised in
that the hollow connecting member (6) is connected to the first end section (12) of the winding element (3) with an interference fit.
3. Conductor according to any one of claims 1 or 2,
characterised in
that the end section (12, 13) has a connecting bore (14, 17) extending from the end face (15) of the winding element (3, 4) to a stop face (16, 18) and having a greater cross section than the channel (7).

4. Conductor according to any one of claims 1 to 3,
characterised in
that the hollow connecting member (6) axially abuts the stop face (16, 18) of
the winding element (3, 4) in the inserted condition.
5. Conductor according to any one of claims 1 to 4,
characterised in
that the hollow connecting member (6) has an outer diameter (D6) that is
greater than the inner diameter (d7) of the channel (7) of the winding element
(3, 4).
6. Conductor according to any one of claims 1 to 5,
characterised in
that the hollow connecting member (6) has a smallest inner diameter (d6) which
is greater than 0.9 times the smallest inner diameter (d7) of the channel (7) of
the winding element (3, 4).
7. Conductor according to any one of claims 1 to 6,
characterised in
that the axial length of the hollow connecting member (6) is at least 0.1 times,
in particular at least 0.2 times the length of the first straight portion (8, 9) of the
winding element (3, 4).
8. Conductor according to any one of claims 1 to 7,
characterised in
that at least one of the hollow connecting member (6) and the winding element
(3, 4) is produced from copper, aluminum, stainless steel or an alloy comprising
at least one of copper, aluminum and stainless steel.
9. Conductor according to any one of claims 1 to 8,
characterised in
that the materials of the hollow connecting member (6) and the winding element
(3, 4) are selected such that a difference between the thermal expansion

coefficient of the material of the hollow connecting member (6) and the thermal expansion coefficient of the material of the winding element (3, 4) is smaller than 10 %.

10. Conductor according to any one of claims 1 to 9,
characterised in
that a second winding element (4) is provided that has the same structure as the first winding element (3),
wherein the first winding element (3) and the second winding element (4) are connected with each other via the hollow connecting member (6) which on one side is axially inserted into the first end section of the first winding element (3) and on the other side is axially inserted into the second end section (13) of the second winding element (4).
11. Conductor according to any one of claims 1 to 10,
characterised in
that the conductor (2) is composed of a plurality of first and second winding elements (3, 4) that are connected to each other via a plurality of hollow connecting members (6), wherein the conductor forms a continuous cooling channel (7) with an inlet (22) and an outlet (23).
12. Conductor according to any one of claims 1 to 9,
characterised in
that the first winding element (3) is connected via the hollow connecting member (6) with a straight intermediate element (11), with one side of the hollow connecting member (6) axially inserted into an end section of the first winding element (3) and the other side of the hollow connecting member (6) axially inserted into the straight intermediate element (11),
and
wherein the straight intermediate element (11) is connected via a second hollow connecting member (6') with a second winding element (4).

13. Electric machine comprising a stator (26) and a rotor (27),

wherein the stator (26) comprises stator core (20) with a plurality of longitudinal slots (21) distributed over the circumference, with a plurality of conductors (2) being arranged in each one of the longitudinal slots (21) of the stator core (20) for generating an electromagnetic field,

characterised in that

at least two of the plurality of conductors (2) is configured according to any one of claims 1 to 12.

14. Electric machine according to claim 13,

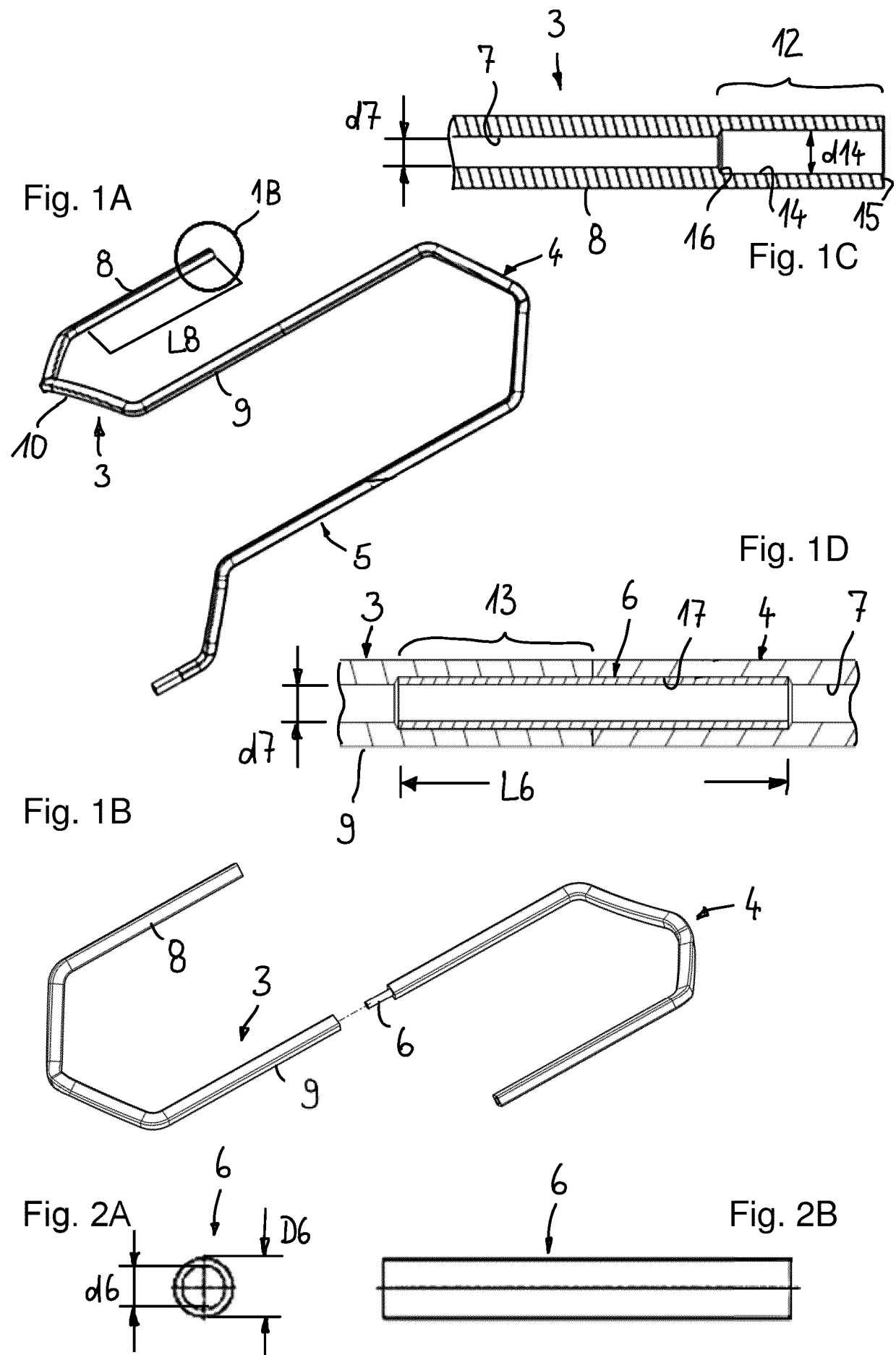
characterised in

that the stator (26) comprises at least four conductor lines in each slot (21), wherein at least two of the plurality of conductor lines are configured to be solid conductors.

15. Electric machine according to claim 13 or 14,

characterised in

that the hollow connecting members (6) of the plurality of conductors (2) are arranged in a center plane arranged between a first end face (32) and an opposite second end face (33) of the stator core (20).



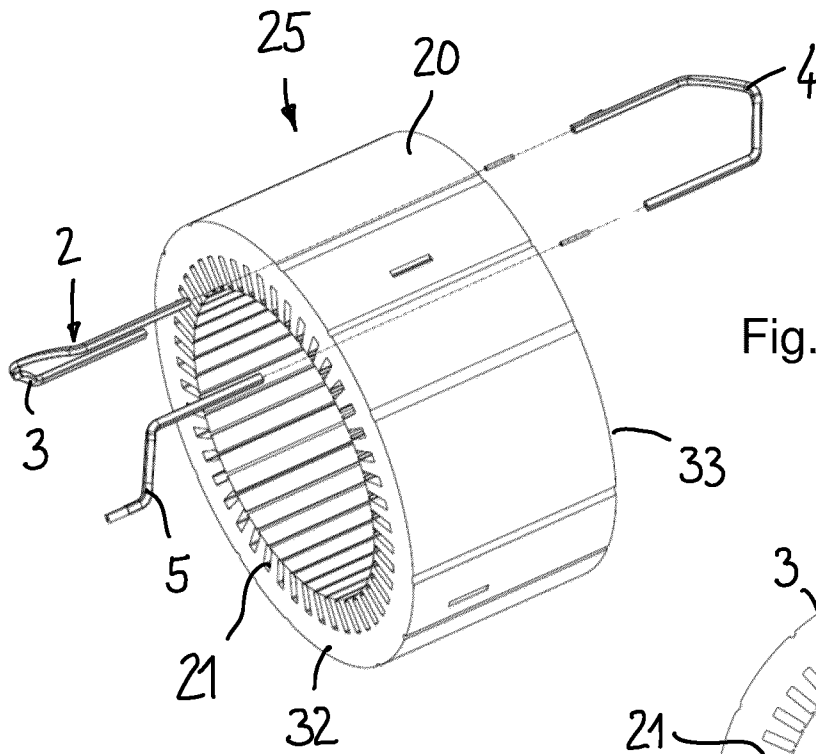


Fig. 3A

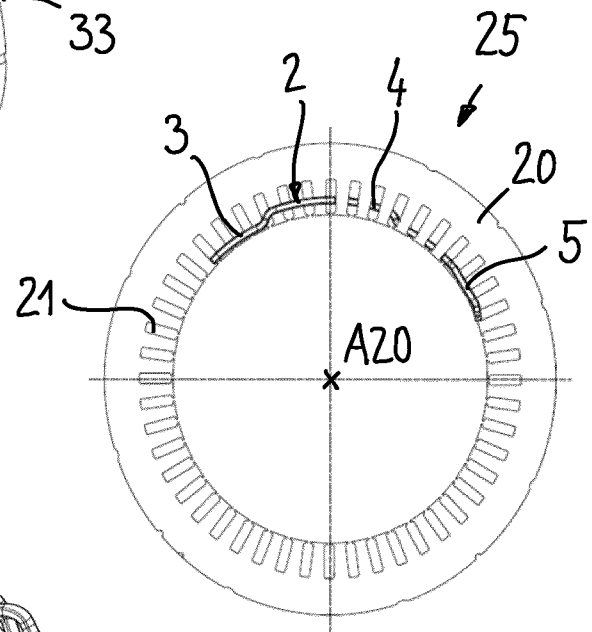


Fig. 3B

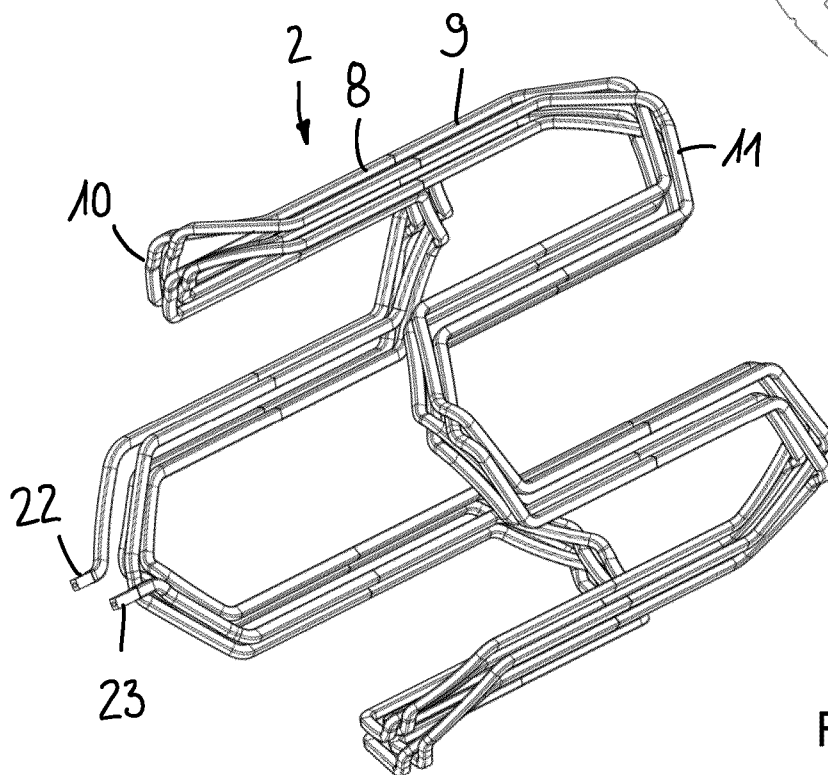


Fig. 4

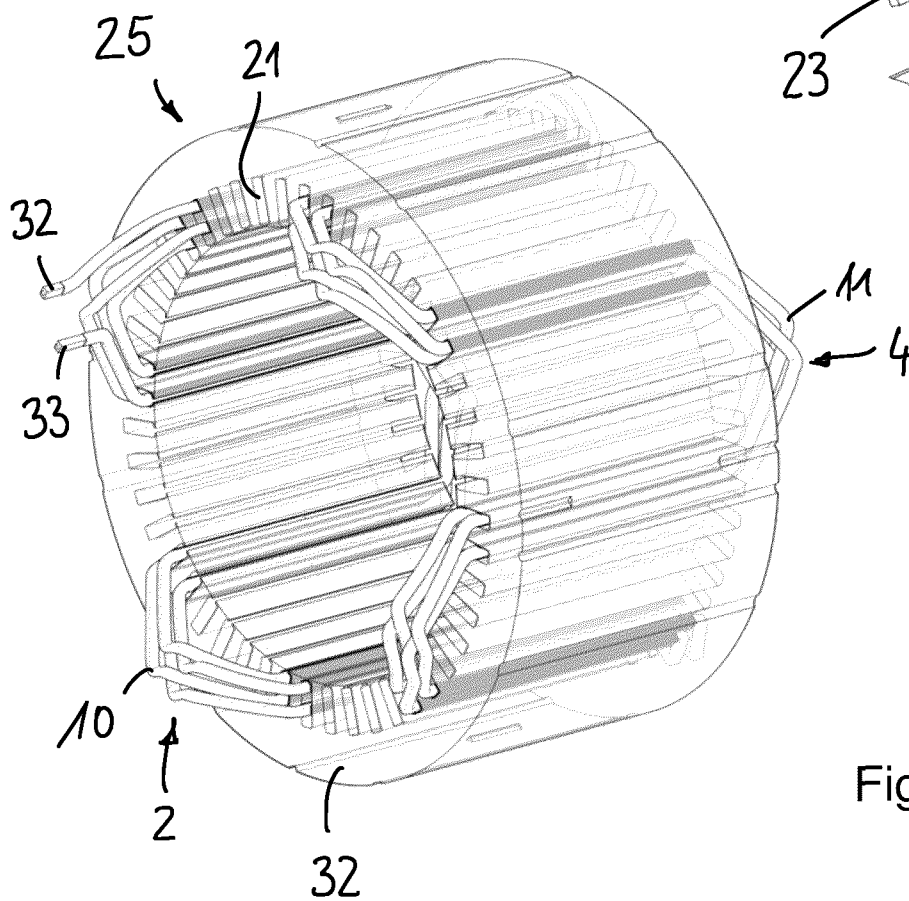
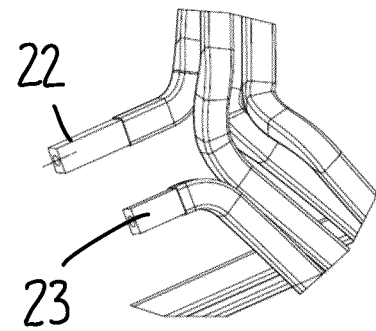
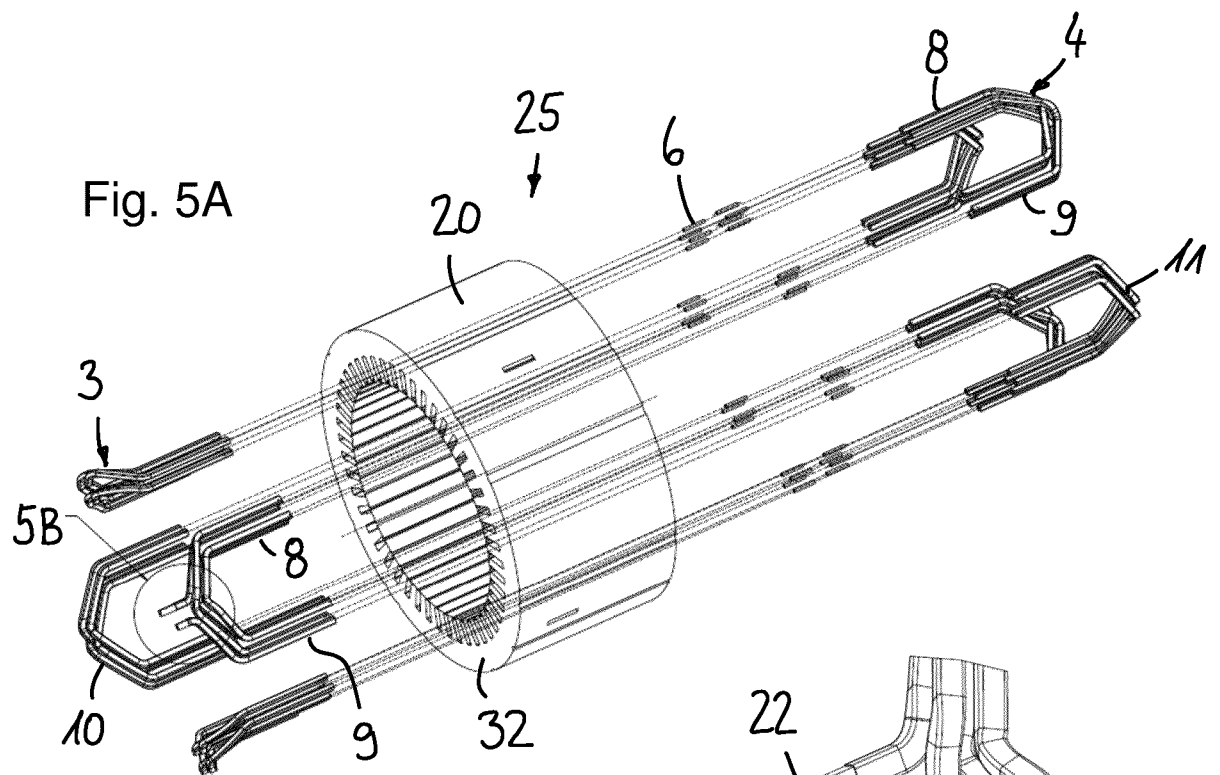


Fig. 5B

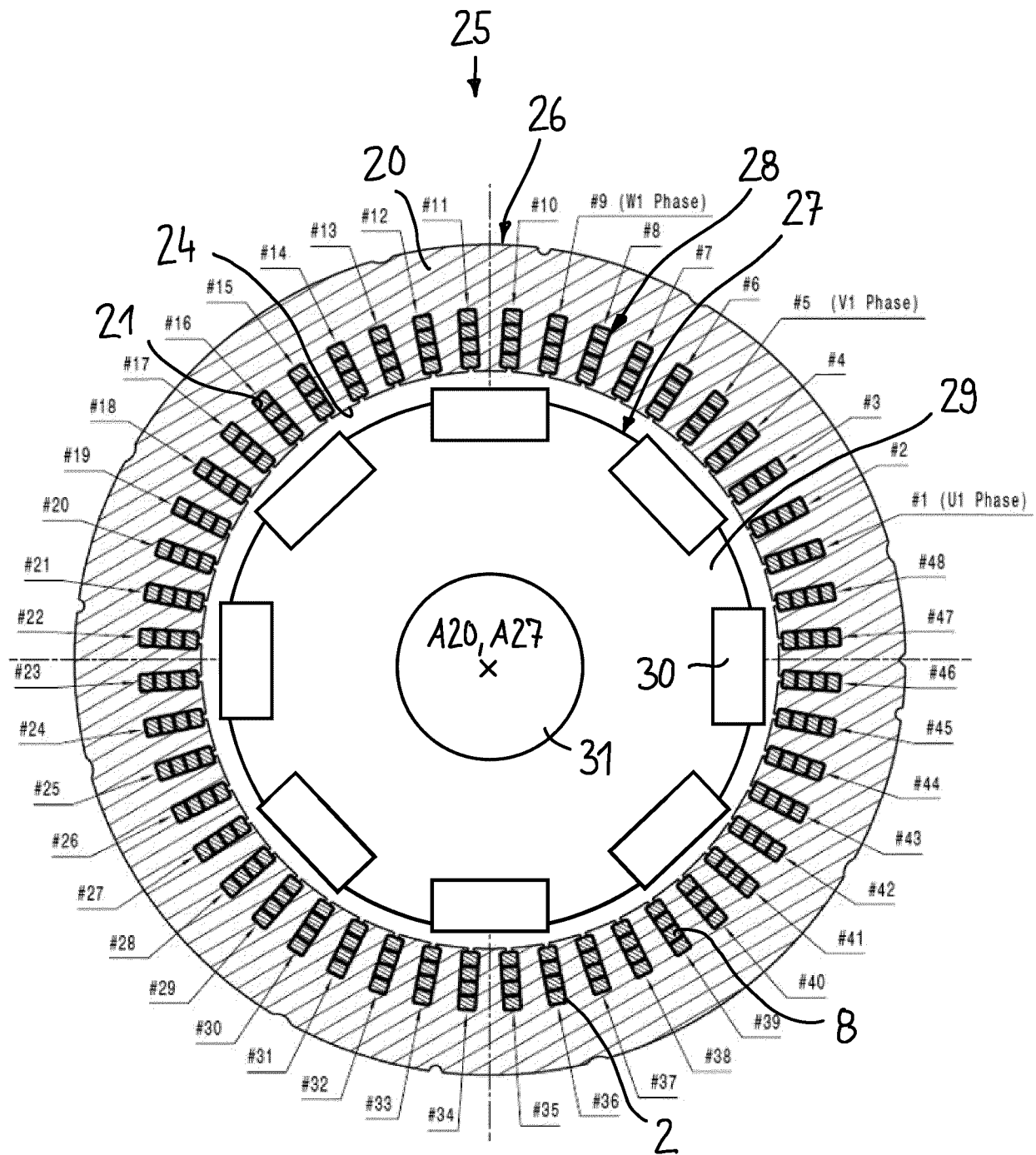


Fig. 6

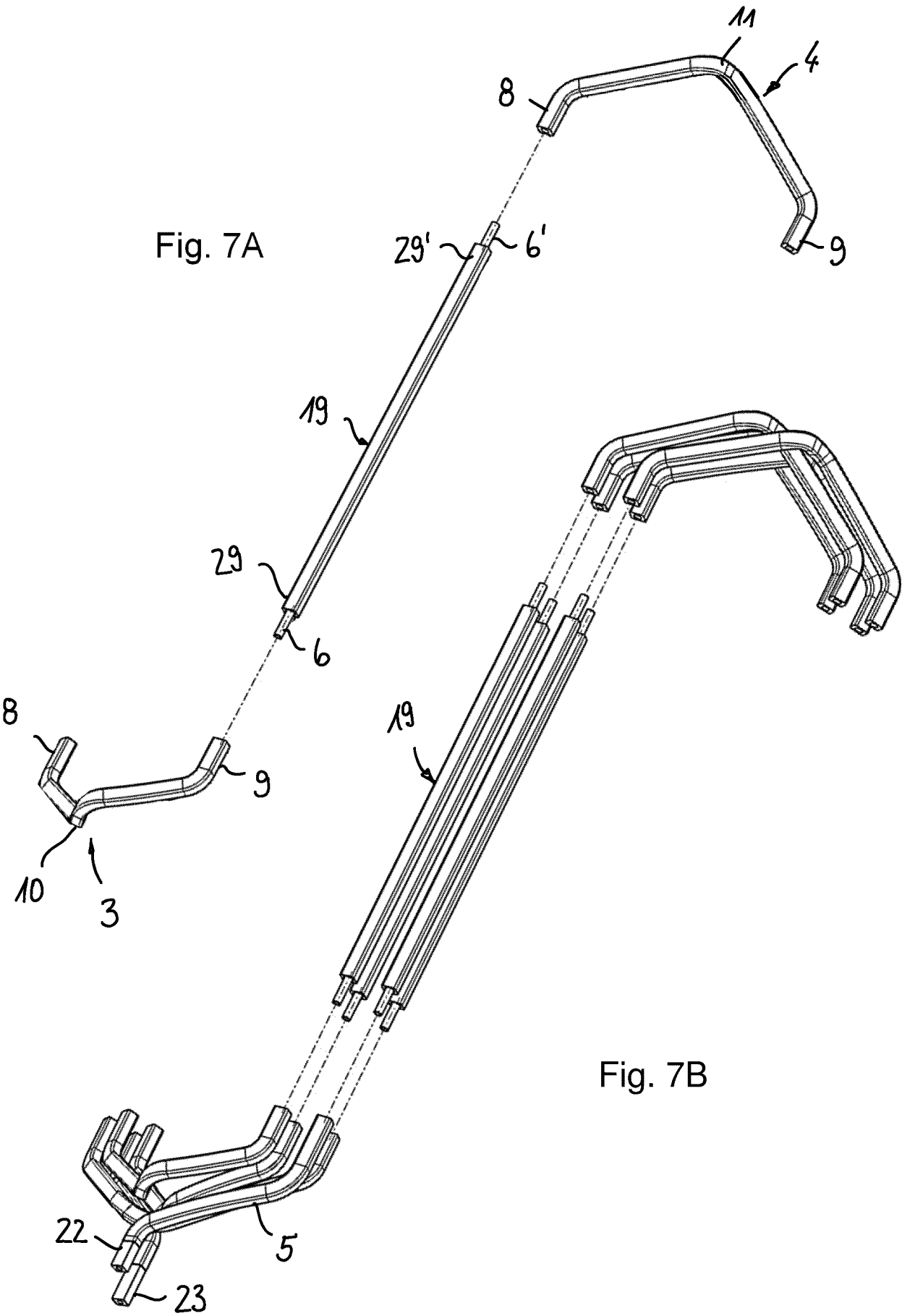


Fig. 7C

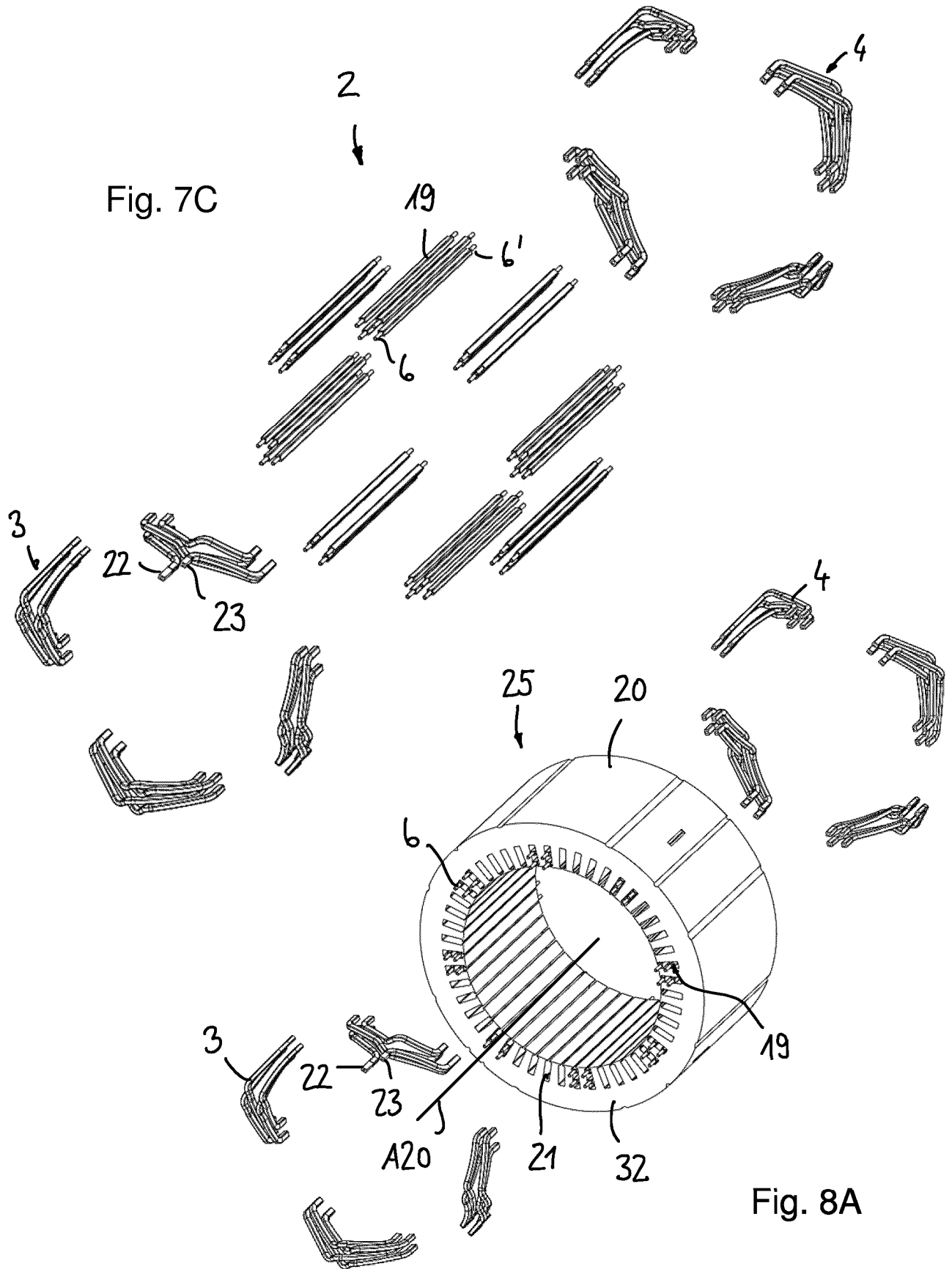
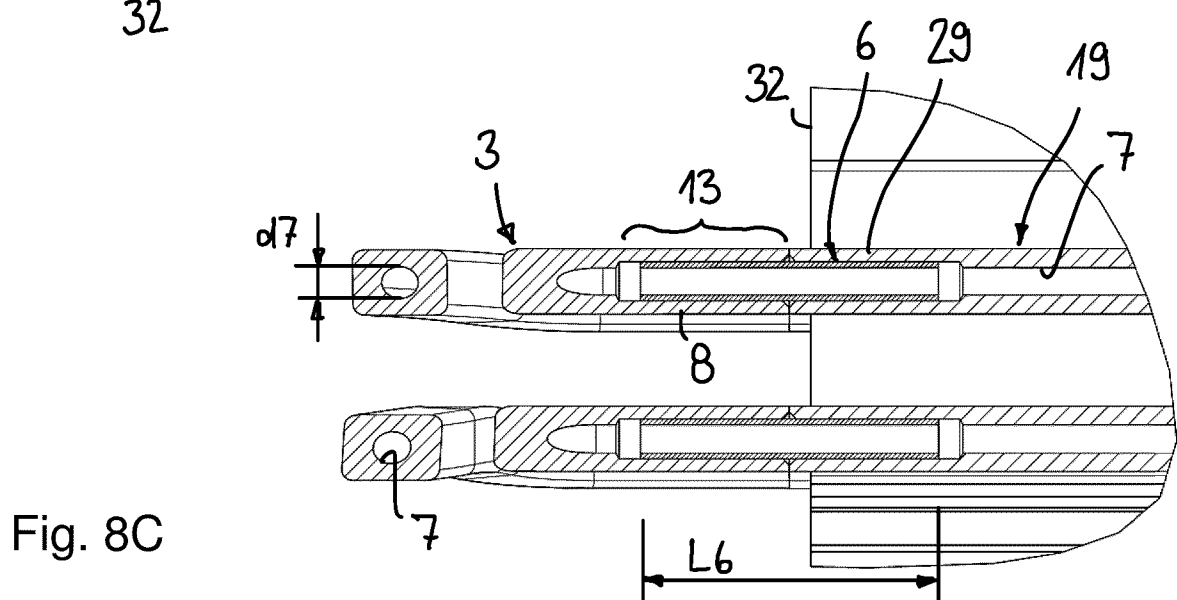
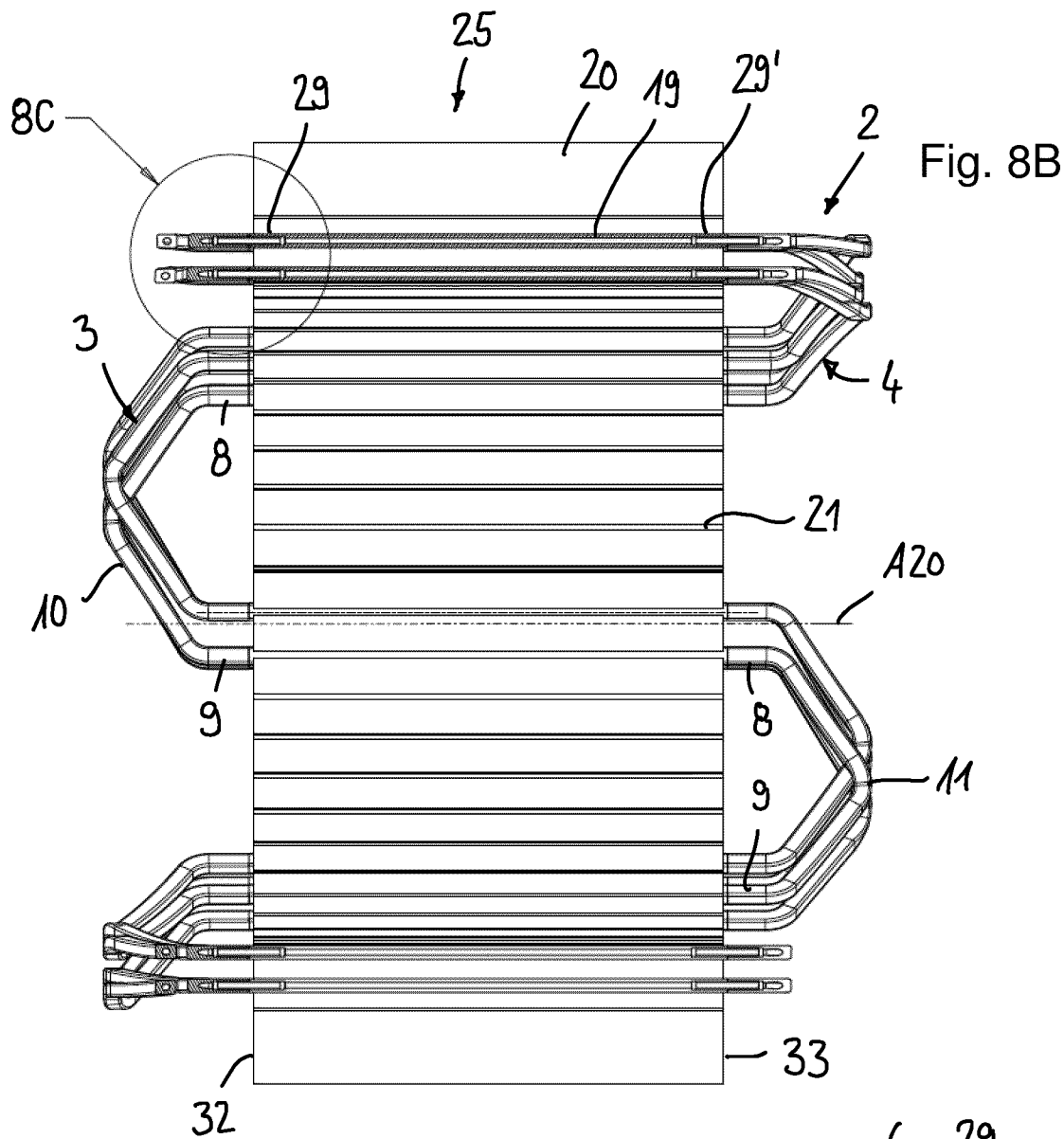


Fig. 8A



INTERNATIONAL SEARCH REPORT

International application No

PCT/EP2022/063071

A. CLASSIFICATION OF SUBJECT MATTER INV. H02K3/12 H02K3/22 H02K3/28 ADD.		
According to International Patent Classification (IPC) or to both national classification and IPC		
B. FIELDS SEARCHED Minimum documentation searched (classification system followed by classification symbols) H02K		
Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched		
Electronic data base consulted during the international search (name of data base and, where practicable, search terms used) EPO-Internal, WPI Data		
C. DOCUMENTS CONSIDERED TO BE RELEVANT		
Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	JP S55 25756 U (TOKYO SHIBAURA ELECTRIC CO) 19 February 1980 (1980-02-19)	1, 2, 9,
Y	paragraph [0002]; figures 5-6	13, 14
	page 3, line 1 - page 5, line 15	3-8,
	-----	10-12, 15
Y	JP S57 80146 U (URAWA MIZOKUCHI TOSHINORI) 18 May 1982 (1982-05-18)	3-8
	figures 1-4	

Y	US 2020/169152 A1 (MATSUMOTO MASASHI [JP] ET AL) 28 May 2020 (2020-05-28)	10-12, 15
	paragraph [0027]; figures 1-4	

A	US 2020/295613 A1 (TOMIOKA TAIZO [JP]) 17 September 2020 (2020-09-17)	14
	figures 1-3	

☐ Further documents are listed in the continuation of Box C. ☒ See patent family annex.		
* Special categories of cited documents : "A" document defining the general state of the art which is not considered to be of particular relevance "E" earlier application or patent but published on or after the international filing date "L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified) "O" document referring to an oral disclosure, use, exhibition or other means "P" document published prior to the international filing date but later than the priority date claimed "T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention "X" document of particular relevance;; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone "Y" document of particular relevance;; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art "&" document member of the same patent family		
Date of the actual completion of the international search		Date of mailing of the international search report
16 December 2022		02/01/2023
Name and mailing address of the ISA/ European Patent Office, P.B. 5818 Patentlaan 2 NL - 2280 HV Rijswijk Tel. (+31-70) 340-2040, Fax: (+31-70) 340-3016		Authorized officer Maître, Jérôme

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

PCT/EP2022/063071

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
JP S5525756	U	19-02-1980	NONE
JP S5780146	U	18-05-1982	NONE
US 2020169152	A1	28-05-2020	CN 111245128 A 05-06-2020
			DE 102019217767 A1 28-05-2020
			JP 2020089119 A 04-06-2020
			US 2020169152 A1 28-05-2020
US 2020295613	A1	17-09-2020	JP 7114513 B2 08-08-2022
			JP 2020150592 A 17-09-2020
			US 2020295613 A1 17-09-2020